

YOLO-Owl: An Occlusion Aware Detector for Low Illuminance Environment

Jiwei Hu*

School of Information Engineering
Wuhan University of Technology
Wuhan, China
*hujiwei@whut.edu.cn

Zhiwei Cui

School of Information Engineering
Wuhan University of Technology
Wuhan, China
2307175441@qq.com

Abstract—Object detection algorithms can be roughly divided into two categories, two-stage detectors like Fast RCNN and one-stage detectors like YOLO. We propose an object detection model that works well in low illuminance environment. We design the MRFM and FEID module for low illuminance targets, as well as a new training model method for low illumination target that allows the model to converge stably and quickly. In addition, we propose an occlusion-aware attention module MPCM to deal with target occlusion in low-light environment. We expand the ExDark dataset and run experiments on the expanded dataset. The results show that the performance of our proposed model is better than YOLOv5 and YOLOX.

Keywords—Object Detection, Low illuminance, Occlusion

I. INTRODUCTION

Common object detection algorithms, such as Faster-RCNN [1], RetinaNet [2], YOLO [3,4,5,6], are rendered ineffective in low illumination scenes due to poor visibility and contrast. Traditional detection methods that first enhance the low illuminance image, and then detect the object do not perform well in dark and blocked scenes. The low illumination object detection algorithm based on domain adaptation proposed by [7] requires the use of paired light and dark data sets to first train the GAN, and then calculate the result by the object detection algorithm. This method's procedure is complex, and the scheme model is difficult to fit, and the experimental conditions are harsh. We propose a network model with relatively simple structure, high accuracy and end-to-end training. In the network's feature extraction section, we design a feature enhancement module that combines dilated convolution and attention mechanism composed of MRFM (Multi receptive field mixer) and FEID (Feature Enhancement in Dark). We are primarily concerned with the occlusion problem between classes. We propose a low light occlusion awareness attention module named MPCM (Multi Patch ConvMixer Model)[8], which is used to emphasize the target area in the image and weaken the background area.

II. RELATED WORKS

The mainstream object detection algorithms based on deep learning are divided into two types: single-stage target detection algorithms and two-stage target detection algorithms. The two-stage algorithm is represented by R-CNN[9], Fast R-CNN[10] and Faster R-CNN. Single stage object detection algorithms represented by SSD series[11] and YOLO series are faster than two-phase detection algorithms in terms of speed. This is due to the fact that such algorithms approach object detection as a regression problem.

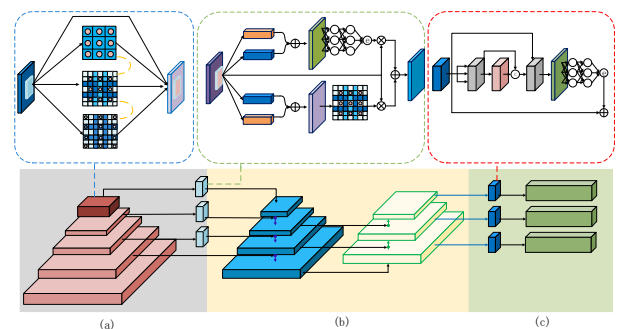
Dilated convolution can increase the receptive field while keeping the feature map size constant. Its essence is an improvement on ordinary convolution by creating some gaps between the pixels of the convolution kernel, allowing the convolution to have a wider range.

Google proposed Mobilenet [12,13,14] in 2017 as a lightweight neural network focusing on mobile terminals or embedded systems. The fundamental unit of the network is the depthwise separable convolution. By splitting the correlation between spatial dimension and channel dimension, the number of parameters required for convolution calculation is reduced, and the use efficiency of convolution kernel parameters is improved.

III. YOLO-OWL

A. Network Architecture

We introduce YOLOv5[15] into low illumination object detection and try to solve the problem of occlusion of the target. Our YOLO-Owl structure is shown in Fig. 1. The backbone is depicted in Figure 1 (a). Based on the overall structure of CSP-Darknet53, we replaced the P5 layer CSP module with MRFM which is shown in the blue dotted line block diagram to obtain a larger receptive field in the low illumination feature map. The green dotted line box depicts the FEID structure, which aims to emphasize the target's key information. MPCM, as shown in the red dotted box, is intended to improve focus on the unobstructed portion of the target.

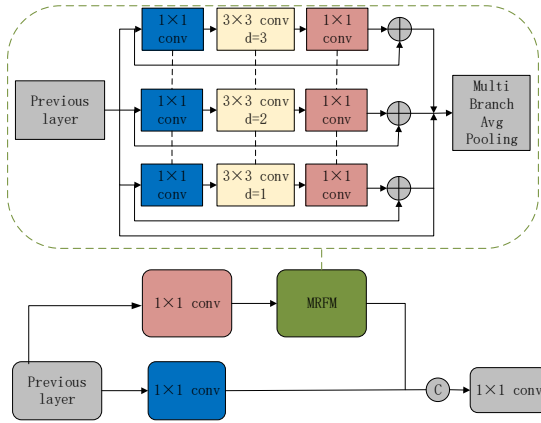


Feature 1. Network architecture of YOLO-Owl. (a) Backbone: Based on the overall structure of CSP-Darknet53. (b) Neck: The overall structure of FPN+PAN has not changed. (c) Head.

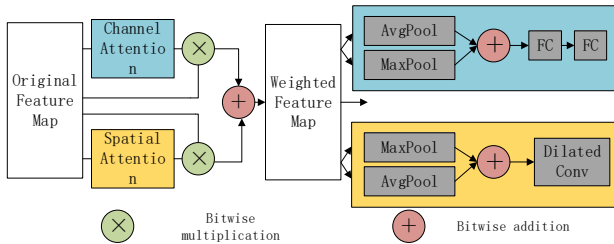
B. Feature Enhancement Model

To fully extract and enhance the target features in low illumination images, we design a feature enhancement module

that combines dilated convolution and attention mechanism. Inspired by TridentNet [16], we design three parallel branch dilated convolution combination modules with different dilation rates, as shown in Fig. 2. The dilated convolution layers with different dilation rates process the same feature map, and then collect the results of different receptive fields by weighting. The convolution parameters of dilation in different branches are shared, which reduces the number of parameters and the potential risk of over fitting. A residual connection is also added to prevent the gradient explosion and disappearance during training. In order to emphasize the key information, the 20×20 , 40×40 and 60×60 feature map which outputted by P5, P4 and P3 layer will respectively enter the key information enhancement module composed of channel spatial attention and external connection attention, as shown in Fig. 3. The original feature map outputs the corresponding weights via the channel attention and spatial attention modules respectively as the weight coefficients of feature map splicing.



Feature 2. Modified CSP block and MRFM module.

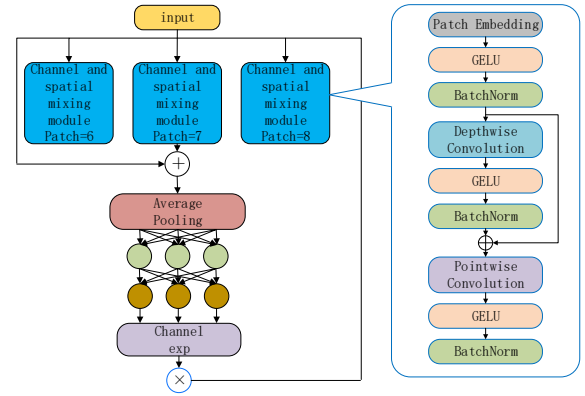


Feature 3. Illustration of FEID. The blue box is the channel attention part, which consists of the maximum pooling layer, the average pooling layer and two full connection layers. The yellow box is the spatial attention part.

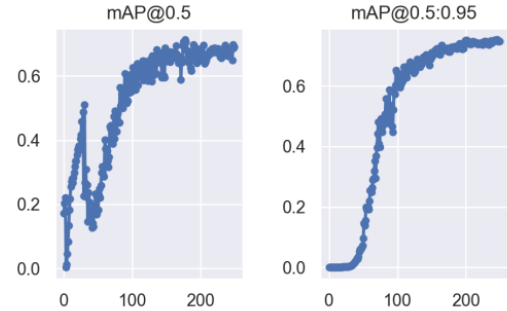
C. Occlusion-Aware Attention Network in Dark

As shown in Fig. 4, we design MPCM to emphasize the target's unobstructed area while weakening the background. The input feature map is split and mapped with three patch sizes to obtain receptive fields at various patch levels and enhance the semantic expression of spatially related information during spatial mapping. Then, we use depthwise separable convolution with residuals. That is, first by a depthwise convolution, then by a pointwise convolution. $M \times K \times K \times 1$ convolution kernels are used to process M channels in

depthwise convolution, learning the importance of different channels channel by channel while reducing the number of parameters and computing cost. To compensate for the missing correlation information between channels in depthwise convolution, $N \times 1 \times 1 \times M$ convolution kernels are used to sum the convolution of the same position of different channels. Then, sum the results of depthwise separable convolution at various patch sizes point by point, while also completing a residual connection to avoid overfitting and gradient disappearance. The value range is then expanded to $[1, e]$ using two full connection layers and exponential normalization to improve the model's robustness to spatial position deviation.



Feature 4. Structure of MPCM.



Feature 5. Training result obtained by using low illumination image directly. The model does not converge well within 250 epochs.

D. A Method of Accelerating Model Convergence

We found labeling an image with extremely low illumination directly difficult in identifying the target and accurately locating the target frame, resulting in difficult convergence of the model or low recognition accuracy, as shown in Fig. 5. Simultaneously, the number of available low illumination data sets of the same type is insufficient, reducing the model's training effect. To address these issues, we propose a feature approximation training method in which we enhance the original low illumination image with Zero DCE[17], label the enhanced image, and then send the label information and the original image into the model training. Simultaneously, we darken the normal illumination image by adding noise to expand the data set for training the model.

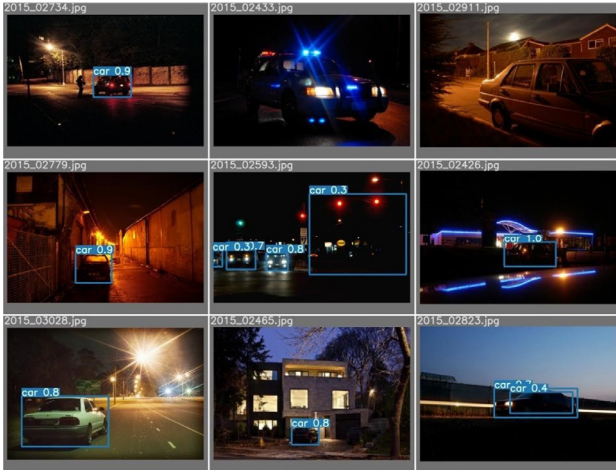
IV. EXPERIMENT

A. Dataset

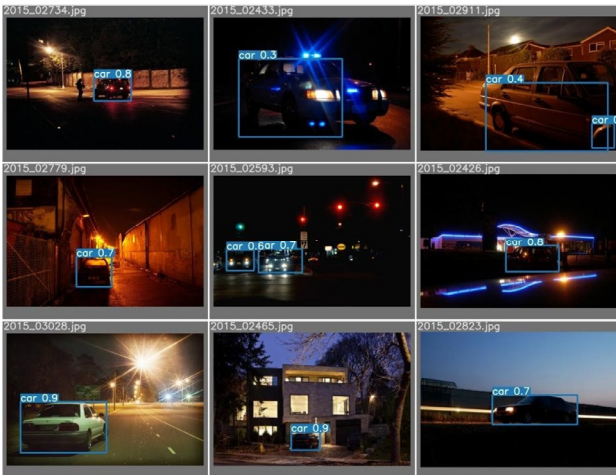
The ExDark dataset [18] is expanded in accordance with our proposed feature approximation solution. We name this data set with E-Exdark (Extended Exclusively Dark). We use the expanded data sets as the target data for experimental comparison. In order to compare the occlusion capability of this model with that of models trained using other methods, we choose 1000 pieces of data from each type of data set for 10%, 20%, and 30% manual occlusion processing. It is used to assess the model's performance in various occluded scenes.

B. Training

All of the training results are obtained using an NVIDIA GeForce RTX 3060 Ti. Each training consisted of a single card training. The epoch was reduced from 250 to 100 based on the results of multiple experiments. The batch size is set to 32. The optimizer we use is SGD with momentum. Because the experiment involves expanding the data set, we redivide the data set. The final training set contains 6547 items, the validation set contains 600 items, and the test set contains 820 items.



Feature 6. Test results of YOLOv5.



Feature 7. Test results of YOLO-Owl.

C. Ablation Study

In this section, we conducted a comprehensive experiment on each module based on the extended E-Exdark dataset. Contrast the mAP of different combinations in the test set with different degrees of occlusion, as shown in Table 1.

Table 1. Ablation study results with different degrees of occlusion.

feature approximation	MRF	FEID	MPC	L0: 0%	L1: 20-40%	L2: 40-60%	L3: 60-80%
✓				75.15	74.80	70.45	65.73
	✓			72.26	70.15	68.70	62.10
		✓		73.40	70.48	68.43	61.37
			✓	73.53	73.20	69.77	64.43
	✓	✓		74.60	73.65	68.75	62.33
✓	✓	✓		76.58	75.98	71.03	66.25
	✓	✓	✓	75.18	74.86	70.86	64.38
✓			✓	76.85	76.78	72.45	67.58
✓	✓	✓	✓	78.18	78.05	75.69	70.42

The experiment shows that in the L0 occlusion degree of low illumination environment, all parts improved the final model's performance. Among these, the feature approximation training method can maximize model accuracy. Each experiment shows good robustness for slight occlusion of L1 degree as the occlusion degree increases. However, when the degree of occlusion was increased to 60-80%, the mAP dropped significantly in all experiments. As shown above, the addition of MPCM modules reduced the mAP the least.

D. Comparisons with Existing Detectors

Table 2. Comparison of our YOLO-Owl and existing detector on the E-Exdark validation dataset.

Detector	Method	mAP(L0)
YOLOv5	YOLOv5	72.15
	EnlightenGAN+YOLOv5	64.30
	KinD+YOLOv5	68.17
	MBLLEN+YOLOv5	73.33
	Zero-DCE+YOLOv5	75.83
YOLOX	YOLOX	75.26
	EnlightenGAN+YOLOX	70.92
	KinD+YOLOX	72.15
	MBLLEN+YOLOX	75.02
	Zero-DCE+YOLOX	75.50
YOLO-Owl		78.18

As shown in Table 2, our method significantly improves the low illumination object detection task. As illustrated in Fig. 6 and Fig. 7, our method outperformed traditional methods in the low-light object detection task.

V. CONCLUSIONS

In this paper, we propose an occlusion aware object detection for illuminance environment based on YOLOv5, called YOLO-Owl. We design the feature enhancement module

of MRFM+FEID, which combines dilated convolution and attention mechanism, to improve the model's extraction of target features and maximize the use of limited feature information of low illumination images. To assist the model in dealing with occlusion, we design a low illumination occlusion perception attention module MPCM. We also propose a method to make the low illumination environment object detection model converge quickly and improve the accuracy. On the E-Exdark dataset, experiments show that our method outperforms YOLOv5, YOLOX, and their combinations with various image enhancement techniques. This paper proposes a method for improving low-light feature extraction, however, how to improve the feature at the input pixel level to better combine the object detection model requires additional research.

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